MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – A)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory.
2. The question paper consists of 11 questions divided into three Sections: A, B, and C.
3. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each.
4. **Section B** comprises 3 Short Answer questions of 2 marks each.
5. **Section C** comprises 3 Long Answer questions of 3 marks each.
6. Use of calculator is not permitted.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. The perimeter of a quadrant of a circle of radius r is: [1]

(A) $\frac{\pi r}{2}$ (B) $2\pi r$ (C) $\frac{\pi r}{2} + 2r$ (D) $\pi r + 2r$

Q2. What is the area of a semi-circle of diameter d? [1]

(A) $\frac{1}{16}\pi d^2$ (B) $\frac{1}{4}\pi d^2$ (C) $\frac{1}{8}\pi d^2$ (D) $\frac{1}{2}\pi d^2$

Q3. The perimeter of a quadrant of a circle of circumference 22 cm is: [1]

(A) 29 cm (B) 22 cm (C) 12.5 cm (D) 15.5 cm

Q4. The area of a sector of a circle of radius 16 cm cut off by an arc of length 18.5 cm is: [1]

(A) 168 cm^2 (B) 148 cm^2 (C) 154 cm^2 (D) 176 cm^2

Q5. **Assertion (A):** If the circumference of a circle is 176 cm, then its radius is 28 cm.
Reason (R): Circumference of a circle $= 2\pi r$, where r is the radius of the circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) (A) is false, but (R) is true.

SECTION B: Short Answer Questions (3 × 2 = 6 Marks)

Q6. The minute hand of a clock is 14 cm long. Find the area on the face of the clock described by the minute hand in 5 minutes. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

Q7. Find the length of the arc of a circle which subtends an angle of 60° at the centre of the circle of radius 42 cm. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

Q8. The perimeter of a sector of a circle of radius 15 cm is 80 cm. Find the area of the sector. [2]

SECTION C: Long Answer Questions (3 × 3 = 9 Marks)

Q9. Find the area of the minor and the major sectors of a circle with radius 6 cm, if the angle subtended by the minor arc at the centre is 60°. [Use $\pi = 3.14$] **[3]**

Q10. A chord of a circle of radius 10 cm subtends an angle of 60° at the centre of the circle. Find the area of the corresponding minor segment of the circle. $\left[\text{Use } \pi = 3.14 \text{ and } \sqrt{3} = 1.73\right]$ **[3]**

Q11. A horse is tied with a rope of length 6 m at the corner of a square grassy lawn of side 20 m. If the length of the rope is increased by 5.5 m (so the new length of rope is 11.5 m), find the increase in the area of the lawn in which the horse can graze. $\left[\text{Use } \pi = \frac{22}{7}\right]$ **[3]**

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★